

# Nicole Ye's Portfolio

[Linkedin](#) | [Nicole Ye's Resume](#) | [My Website!](#) | [n7ye@uwaterloo.ca](mailto:n7ye@uwaterloo.ca) | [Github](#)

If you have any inquiries, please don't hesitate to reach out!

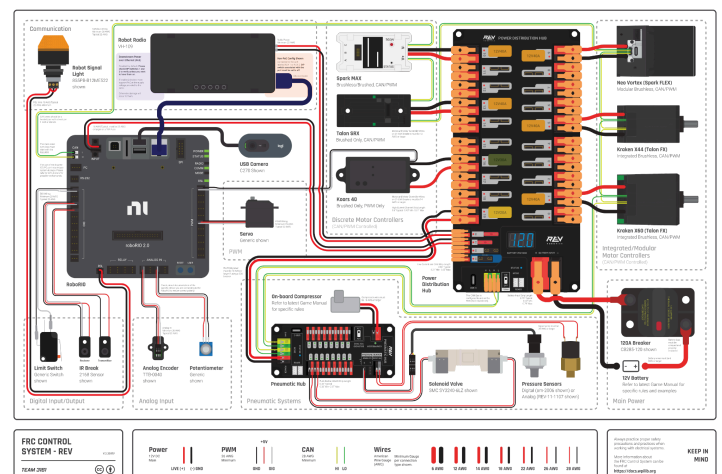
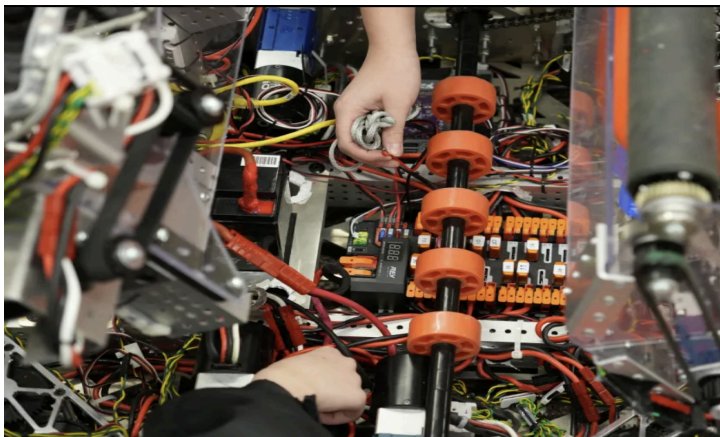
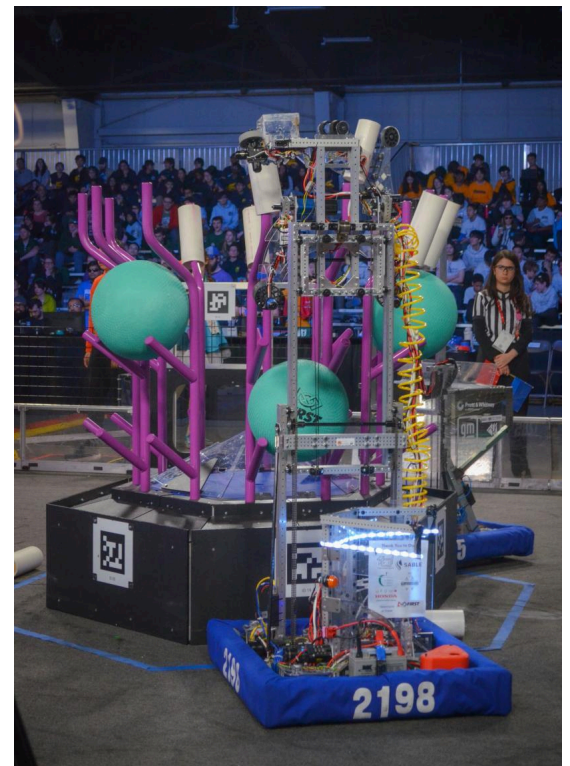
## SKILLS

- **Languages:** Java, Python, Javascript, HTML/CSS, C++
- **Programs:** Onshape, Solidworks, Visual Studio Code, Microsoft Excel/Teams, Slack, Discord
- **Electrical Skills:** Soldering, Multimeter, Cable Management, Crimping, Testing
- **Soft Skills:** Project Management, Critical Thinking, Organization, Customer Service

## Team 2198 Paradigm Shift Robotics Electrical System

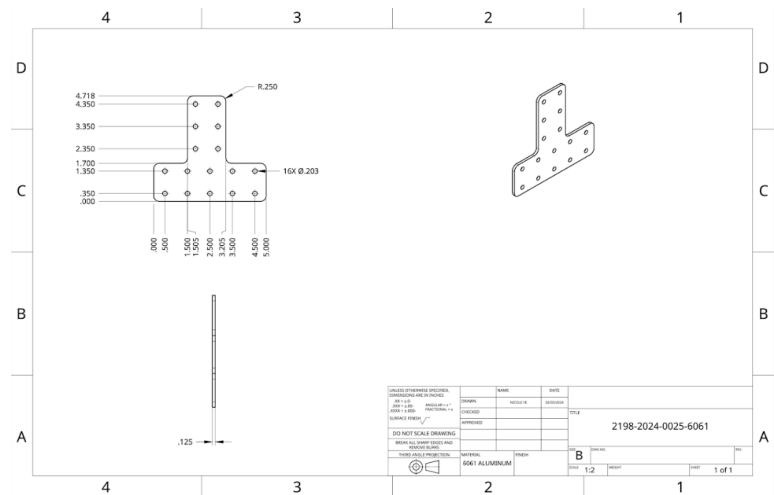
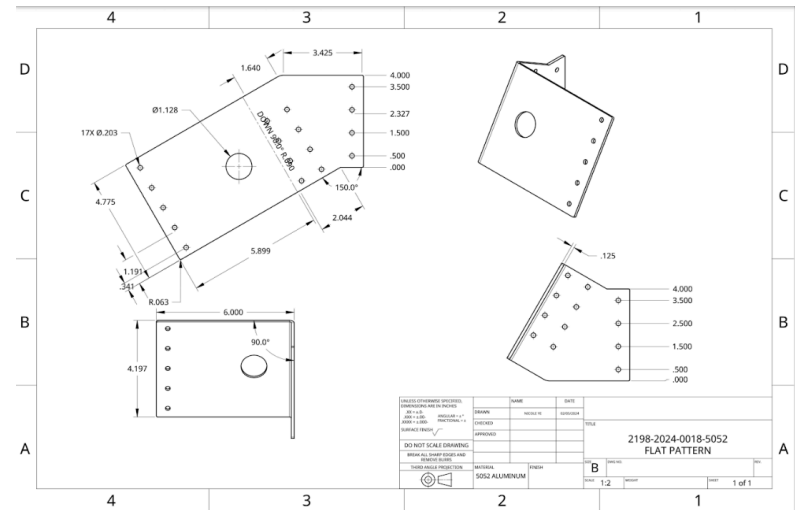
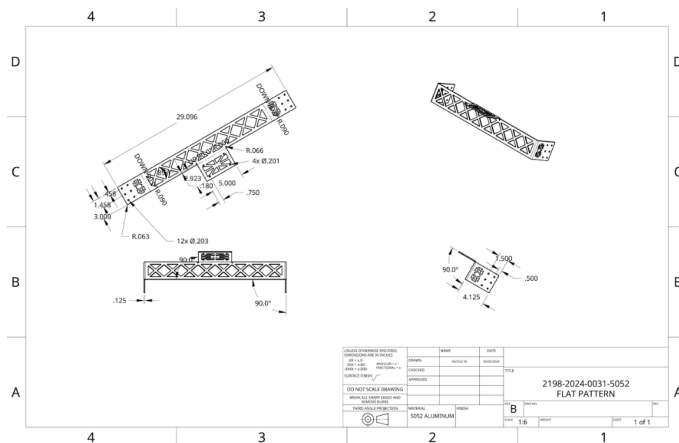
Sep. 2021 - June 2025

- Utilized [TKO Circuit Simulator](#) to design the layout of electrical components 2024-2025 robot
- Designed and implemented a customized electrical system, including wiring, power distribution, and sensor integration, ensuring compliance with competition standards
- Integrated a **line sensor** to detect whether a game piece was secured in the holding mechanism, improving automation
- Wired **encoders** for precise motor control and accurate movement tracking across drive and manipulator systems
- Implemented **color sensors** to differentiate game piece types and enable autonomous decision-making
- Mentored and trained **10+ new members** in electrical wiring, sensor integration, and troubleshooting best practices
- Collaborated with mechanical and programming teams to ensure seamless integration between electrical subsystems



# Robot Part Drawings OnShape/Solidworks

- Utilized Solidworks and Onshape to create DXF, STEP, and PDF drawings of sheet metal of First Robotics Robots (2021-2025) for CNC processing
- Designed and fabricated gussets for a FIRST Robotics robot, ensuring precise alignment and secure connections between major mechanisms to enhance structural integrity
- Followed set of tools utilizing standard views, section, and detail dimensions such as model items and smart Dimension, and annotations (including center marks and bill of materials)
- Followed up with metal sponsor with metal parts list organized by part number in folders using Excel



# Ecommerce System / Python

- Designed a flowchart to represent the control flow and decision-making process of program
- Integrated key programming concepts like conditional statements, arrays, and math operators to create product sorting, search functionality, and checkout
- Utilized arrays (lists) to store and manage dynamic product data including names and categories
- Implemented bubble sort algorithm to organize product listings by price and rating for user experience
- Designed structured test cases to evaluate different input scenarios and confirm expected program outputs

```

Please search below.
Search: fountain pen
Results Of Search

Item Number - 1
Name: wordsworth & black fountain pen set, medium nib
Description: includes 6 ink cartridges and ink refill converter pens [silver gold]
Price: $ 25.0
Rating: 5 / 5

Item Number - 2
Name: jinhao x450 fountain pen (green marble, medium nib 0.7mm)
Description: nib: m (steel nib), cap: push type; converter: screw type
Price: $ 25.5
Rating: 5 / 5

Item Number - 3
Name: arteza - 140-pcs wooden drawing kit
Description: includes 60 crayons, 60 watercolour pans, 10 paintbrushes, 10 hb pencils
Price: $ 40.99
Rating: 5 / 5

Item Number - 4
Name: super squiggles markers - 12 colors self outline markers
Description: double line pen, outline markers pens for art
Price: $ 60.99
Rating: 5 / 5

Item Number - 5
Name: wordsworth & black fountain pen set, medium nib
Description: includes 6 ink cartridges and ink refill converter pens [silver gold]
Price: $ 25.0
Rating: 5 / 5

Would you like to sort items by price, add item to cart, or search for a different item?
1) Enter 1 to sort items by price
2) Enter 2 to add item to cart
3) Enter 3 to Search for a Different Product
Answer: []
    
```

# Tutoring Service - Space Academy / JavaScript/HTML/CSS

- Utilized JavaScript to manage cart functionality and total cost calculations
- Created interactive buttons to select courses, add them to a shopping cart, and proceed to checkout
- Designed a clean responsive webpage using HTML and CSS, ensuring a seamless user experience.



## CAT'S SPACE ACADEMY

About Us

All Aboard Cat's Space Academy! Whether you're seeking help with math or science, we have all the courses you need. As well, there's a little secret, if you select the 6 pack (All Gr.12 Maths and Sciences), you will receive a 20% discount on your overall purchase! Now click the buttons to select your courses, so I can take off with your money oops I mean to space!:)

My Very Cheap Tutoring Pricing:

|                  |         |
|------------------|---------|
| Grade 9 Courses  | \$30/hr |
| Grade 10 Courses | \$40/hr |
| Grade 11 Courses | \$45/hr |
| Grade 12 Courses | \$50/hr |

| Math                                                                                                                                                           | Science                                                                                                                                                                            | Cart                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |             |           |              |      |            |      |            |      |            |      |                         |      |                       |      |                    |      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------|--------------|------|------------|------|------------|------|------------|------|-------------------------|------|-----------------------|------|--------------------|------|
| <div>Math 9</div> <div>Math 10</div> <div>Functions 11</div> <div>Data Management 12</div> <div>Advanced Functions 12</div> <div>Calculus and Vectors 12</div> | <div>Science 9</div> <div>Science 10</div> <div>Chemistry 11</div> <div>Biology 11</div> <div>Physics 11</div> <div>Chemistry 12</div> <div>Biology 12</div> <div>Physics 12</div> | <table><thead><tr><th>&lt; Subject &gt;</th><th>&lt; Price &gt;</th></tr></thead><tbody><tr><td>Chemistry 12</td><td>\$50</td></tr><tr><td>Physics 11</td><td>\$45</td></tr><tr><td>Biology 12</td><td>\$50</td></tr><tr><td>Physics 12</td><td>\$50</td></tr><tr><td>Calculus and Vectors 12</td><td>\$50</td></tr><tr><td>Advanced Functions 12</td><td>\$50</td></tr><tr><td>Data Management 12</td><td>\$50</td></tr></tbody></table> <p>6 Pack Applied: 20% Discount!</p> <p>Subtotal: \$276</p> <p>Tax (13%): \$ 35.88</p> <p>Total: \$311.88</p> | < Subject > | < Price > | Chemistry 12 | \$50 | Physics 11 | \$45 | Biology 12 | \$50 | Physics 12 | \$50 | Calculus and Vectors 12 | \$50 | Advanced Functions 12 | \$50 | Data Management 12 | \$50 |
| < Subject >                                                                                                                                                    | < Price >                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |           |              |      |            |      |            |      |            |      |                         |      |                       |      |                    |      |
| Chemistry 12                                                                                                                                                   | \$50                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |           |              |      |            |      |            |      |            |      |                         |      |                       |      |                    |      |
| Physics 11                                                                                                                                                     | \$45                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |           |              |      |            |      |            |      |            |      |                         |      |                       |      |                    |      |
| Biology 12                                                                                                                                                     | \$50                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |           |              |      |            |      |            |      |            |      |                         |      |                       |      |                    |      |
| Physics 12                                                                                                                                                     | \$50                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |           |              |      |            |      |            |      |            |      |                         |      |                       |      |                    |      |
| Calculus and Vectors 12                                                                                                                                        | \$50                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |           |              |      |            |      |            |      |            |      |                         |      |                       |      |                    |      |
| Advanced Functions 12                                                                                                                                          | \$50                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |           |              |      |            |      |            |      |            |      |                         |      |                       |      |                    |      |
| Data Management 12                                                                                                                                             | \$50                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |           |              |      |            |      |            |      |            |      |                         |      |                       |      |                    |      |